SAIKET SYSTEMS · BUSINESS ANALYSIS INTERNSHIP · PROJECT REPORT

Customer Churn Analysis

in a Telecommunications Company

TASKS 1 – 4

7,043	26.54%	11	3
CUSTOMERS ANALYSED	OVERALL CHURN RATE	HIDDEN MISSING VALUES FIXED	LIFECYCLE SEGMENTS

Prepared by	FEDELIKA MAXMUS
Programme	Saiket Systems — Business Analysis Program
Dataset	Telco Customer Churn — 7,043 customers × 21 attributes

Dataset Understanding · Data Cleaning · Exploratory Data Analysis · Customer Segmentation Visualization

Contents

1. Executive Summary 4

1.1 Project objective . . . 4
1.2 Headline findings . . . 4
1.3 What a business reader should take away . . . 4

2. Project Overview & Dataset 5

2.1 Scope of this report . . . 5
2.2 The dataset . . . 5
2.3 Tools used . . . 5

3. Task 1 — Understand the Dataset 6

3.1 Steps performed . . . 6
3.2 Structure and first records . . . 6
3.3 Column data types . . . 6
3.4 Missing values — the check that mattered . . . 7
3.5 Findings . . . 7

4. Task 2 — Data Cleaning 8

4.1 Cleaning plan . . . 8
4.2 Step 1 — Standardising column names . . . 8
4.3 Step 2 — Diagnosing and handling missing values . . . 8
4.4 Step 3 — Duplicate removal . . . 9
4.5 Final hygiene and validation . . . 10
4.6 Findings . . . 10

5. Task 3 — Exploratory Data Analysis 11

5.1 Summary statistics . . . 11
5.2 Distributions of the numerical columns . . . 11
5.3 Spread and outliers, split by churn . . . 12
5.4 Churn rate — overall . . . 12
5.5 Where the churn sits — breakdowns by contract, service and payment . . . 13
5.6 Findings . . . 14

6. Task 4 — Customer Segmentation Visualization 14

6.1 How the segments were defined . . . 14

6.2 Customer distribution by tenure — donut chart . . . 14
6.3 Average monthly charges by tenure — clustered bar chart . . . 16
6.4 Findings . . . 17

7. Consolidated Findings & Recommendations **17**

7.1 Findings . . . 17
7.2 Recommended actions . . . 18
7.3 Measurement . . . 18

8. Conclusion **19**

Appendix A. Reproducibility Notes **19**

A.1 How to reproduce this report . . . 19
A.2 Environment . . . 19
A.3 File inventory . . . 20
A.4 Validation summary . . . 20

List of Figures

Figure 1. Distributions of tenure, monthly charges and total charges 12
Figure 2. Box plots of each numerical column, split by churn 12
Figure 3. Churn counts and proportions 13
Figure 4. Churn composition by contract, service and payment method 14
Figure 5. Customer distribution by tenure segment (Matplotlib donut) 15
Figure 6. Customer distribution by tenure (interactive Plotly donut) 15
Figure 7. Average monthly charges by tenure, clustered by churn (Matplotlib) 16
Figure 8. Average monthly charges by tenure (interactive Plotly) 16

1. Executive Summary

What was analysed, what was found, and what a business reader should do with it.

7,043	26.54%	11	0	3
CUSTOMERS ANALYSED	OVERALL CHURN RATE	HIDDEN MISSING VALUES FOUND	ROWS LOST IN CLEANING	LIFECYCLE SEGMENTS BUILT

1.1 Project objective

This report documents the first four tasks of the Business Analysis internship project on customer churn in a telecommunications company. The wider project goal is to reduce churn and improve customer retention; the immediate goal of Tasks 1–4 is to build a dataset that can be trusted, understand what the data says, and segment the customer base so that later predictive modelling and retention campaigns can be aimed at the right people.

Every figure quoted in this report was produced by the accompanying Jupyter notebooks and re-verified against the cleaned dataset before publication.

1.2 Headline findings

- **The headline problem is the first year.** Customers in their first 12 months churn at **47.44%**, versus **11.93%** for customers past 37 months — a four-fold difference. Newcomers are also the largest single risk pool: 2,186 customers, 31.04% of the base.
- **Churners are not cheap customers — they are expensive ones who leave early.** They pay **$74.44** per month against **$61.27** for those who stay (+21.5%), yet their average tenure is only **17.98 months** against **37.57 months**. Lifetime value is therefore roughly a third lower ($1,531.80 vs $2,549.91).
- **Value rises with tenure, so early retention compounds.** Average monthly charges climb **$56.10 → $63.25 → $72.01** across the three lifecycle segments (+28.4% from newcomer to loyal). Saving a customer in year one is worth more than saving the same customer in year five.
- **Contract type is the strongest commercial lever in the data.** Month-to-month churns at **42.71%** against **11.27%** (one year) and **2.83%** (two year) — a 15× gap.
- **Payment friction is a leading indicator.** Electronic check churns at **45.29%** versus **15–17%** for the two automatic methods.
- **Product mix matters.** Fiber optic churns at **41.89%** versus DSL **18.96%** and no-internet customers **7.40%**.
- **The raw file was quietly dirty.** isnull() reported zero missing values, but 11 totalcharges cells held blank strings. These were identified, justified and repaired in Task 2 without discarding a single customer.

1.3 What a business reader should take away

> **The one-line summary**
>
> A quarter of this customer base leaves every year, and almost all of the leakage happens before a customer ever reaches their first anniversary. The customers who leave are the ones paying the most, which means the loss is not only in headcount but in high-bill revenue. The controllable levers visible in the data are contract structure, payment method and early-life experience — all three are commercial decisions, not model decisions.

2. Project Overview & Dataset

Scope of work, the data being analysed, and how each task maps to the brief.

2.1 Scope of this report

The internship brief asks for four of five tasks to be completed as separate deliverables. This report covers **Tasks 1 to 4**. Task 5 (Advanced Analysis) is delivered separately and builds directly on the cleaned dataset and the tenure segments defined here.

Task	Brief requirement	Status
1. Understand the Dataset	Load with pandas, show first 10 rows, identify data types, check for missing values	Complete
2. Data Cleaning	Handle missing values, remove duplicates, standardise column names to lowercase with underscores	Complete
3. Exploratory Data Analysis	Summary statistics (mean/median/mode), histograms and box plots, churn vs non-churn analysis	Complete
4. Customer Segmentation Visualization	Pie/donut of customers by tenure, clustered bar of average monthly charges by tenure with annotations, interactive visualisation	Complete

2.2 The dataset

The source file is a customer-level telecom churn dataset: one row per customer, one column per attribute, plus a binary Churn target indicating whether that customer cancelled during the observation window.

Attribute	Value
Source file	Telco_Customer_Churn_Dataset (3).csv
Working file (from Task 2)	Telco_Customer_Churn_Cleaned.csv
Rows (customers)	7,043
Columns (attributes + target)	21 (customerid + 19 features + churn)
Grain	One row per customer; customerid is unique (7,043 / 7,043)
Numeric features	tenure (months), monthlycharges ($), totalcharges ($), seniorcitizen (0/1)
Categorical features	gender, partner, dependents, 10 service flags, contract, paperlessbilling, paymentmethod
Target variable	churn — No (5,174 / 73.46%) vs Yes (1,869 / 26.54%)
Observation window	Tenure spans 0–72 months (max contract length)

2.3 Tools used

A single consistent Python stack is used across all four notebooks so that results are directly comparable between tasks. Interactive charts required by the brief are produced with Plotly; the static versions in this report are rendered from the same code.

Library	Version	Role in this project
Python	3.14.7	Runtime for all four notebooks
pandas	3.0.6	Loading, cleaning, grouping, crosstabs

Library	Version	Role in this project
NumPy	2.5.3	Numeric operations and chart positioning
Matplotlib	3.11.2	Static histograms, box plots, donut and clustered bars
Seaborn	0.13.2	Statistical chart styling (histograms, box plots, count plots)
Plotly	7.1.0	Interactive donut and clustered bar (brief requirement)
Jupyter / JupyterLab	—	Notebook execution environment for all deliverables

3. Task 1 — Understand the Dataset

Objective: load the dataset with pandas, inspect its structure, and identify the data issues that later tasks must fix.

3.1 Steps performed

- Loaded the raw CSV into a pandas DataFrame and recorded its shape and column list.
- Displayed the first 10 rows to see real values and spot formatting problems.
- Inspected the data type of every column using dtypes and info().
- Computed describe() for the numerical columns.
- Checked for missing values with isnull() — and then tested for hidden blanks, which turned out to matter.
- Checked for duplicate rows and verified that customerid is unique.

3.2 Structure and first records

The dataset is **7,043 rows × 21 columns**. The first five records are shown below; the full first-10-row view is produced in the notebook.

customerID	gender	SeniorCitizen	tenure	Contract	MonthlyCharges	TotalCharges	Churn
7590-VHVEG	Female	0	1	Month-to-month	29.85	29.85	No
5575-GNVDE	Male	0	34	One year	56.95	1889.50	No
3668-QPYBK	Male	0	2	Month-to-month	53.85	108.15	Yes
7795-CFOCW	Male	0	45	One year	42.30	1840.75	No
9237-HQITU	Female	0	2	Month-to-month	70.70	151.65	Yes

3.3 Column data types

Data type	Count	Columns
object / string	18	customerid, gender, partner, dependents, phoneservice, multiplelines, internetservice, onlinesecurity, onlinebackup, deviceprotection, techsupport, streamingtv, streamingmovies, contract, paperlessbilling, paymentmethod, **totalcharges**, churn
int64	2	seniorcitizen, tenure
float64	1	monthlycharges

Issue spotted at this stage

TotalCharges is stored as text, not as a number. It cannot be averaged or plotted until it is coerced. This is exactly the kind of defect that Task 1 exists to surface, and it is resolved in Task 2.

3.4 Missing values — the check that mattered

The standard check reports a clean dataset. A deeper check does not:

Check	Result	Interpretation
df.isnull().sum().sum()	0	Appears complete — but this is misleading
Blank / whitespace strings per text column	11 in TotalCharges, 0 in all 17 others	Missing values exist but are stored as empty strings
pd.to_numeric(TotalCharges, errors='coerce')	11 NaN	Confirms 11 genuine missing values once blanks become real nulls
Tenure of the 11 affected rows	tenure = 0 for all 11	Brand-new customers with no billing history yet
df.duplicated().sum()	0	No duplicate rows
customerid uniqueness	7,043 / 7,043	No ID collisions

Why this matters

A cleaning script written only against isnull() would have declared this dataset perfect and carried 11 empty strings silently into every average, chart and model built on top of it. The 11 affected rows are only 0.16% of the data, but they sit at the extreme low end of lifetime value and would visibly distort the totalcharges distribution.

3.5 Findings

- **Size and grain:** 7,043 unique customers × 21 attributes; one row per customer.
- **Target:** Churn = Yes for 1,869 customers (26.54%) — a substantial minority, so the dataset is usable for analysis without resampling.
- **Types:** only three attributes are genuinely numeric (tenure, MonthlyCharges, SeniorCitizen). TotalCharges is text and must be coerced.
- **Missing data:** zero true nulls, but **11 hidden blanks** in TotalCharges, all belonging to customers with tenure = 0.
- **Duplicates:** none, and the business key is unique.
- **Modal customer profile:** male, no partner, no dependents, month-to-month contract, fiber optic internet, electronic check payment, did not churn.
- **Handed to Task 2:** coerce TotalCharges to numeric, resolve the 11 blanks, standardise column naming, confirm no duplicates.

4. Task 2 — Data Cleaning

Objective: make the dataset safe to analyse — handle missing values, remove duplicates and standardise column names.

4.1 Cleaning plan

Task 1 handed over three concrete defects. Each is addressed by an explicit, justified step, and a before/after snapshot is printed so the effect of every change is auditable.

#	Defect found in Task 1	Action taken in Task 2
1	Mixed-case, inconsistent column names (customerID, SeniorCitizen, TotalCharges)	Lowercase + spaces to underscores + strip, then verified with assertions
2	TotalCharges stored as text	Coerced to numeric with pd.to_numeric(errors='coerce') → float64
3	11 hidden blank values in TotalCharges	Diagnosed, a fill strategy chosen on business grounds, and filled with 0
4	Duplicates (none found, but unverified)	duplicated() counted, customerid uniqueness proven, drop_duplicates() applied idempotently

4.2 Step 1 — Standardising column names

The brief requires lowercase names with spaces replaced by underscores. This is not cosmetic: SeniorCitizen versus seniorcitizen versus Senior Citizen produces silent KeyErrors and broken groupings in every downstream task. The rename is therefore validated with three assertions — all lowercase, no spaces, no duplicate names.

```
df.columns = df.columns.str.strip().str.lower().str.replace(" ", "_", regex=False)

assert (df.columns == [c.lower() for c in df.columns]).all()   # all lowercase
assert not any(" " in c for c in df.columns)                   # no spaces
assert not df.columns.duplicated().any()                        # no duplicate names
```

Before	After
customerID	customerid
SeniorCitizen	seniorcitizen
MonthlyCharges	monthlycharges
TotalCharges	totalcharges
Churn	churn
… 21 columns in total	… 21 columns, all lowercase, no spaces

4.3 Step 2 — Diagnosing and handling missing values

Missing values were searched for in three layers, because a single isnull() call had already given a false all-clear: standard null counts; blank and whitespace-only strings per text column; and numeric coercion with errors='coerce', which converts blanks into real nulls that can be counted.

```
# Layer 1 — standard nulls
df.isnull().sum().sum()                      # -> 0   (misleading)

# Layer 2 — blank / whitespace-only strings
{c: int((df[c].astype(str).str.strip() == "").sum())
 for c in df.select_dtypes(include=["object", "string"]).columns}
# -> {'totalcharges': 11, ...all others 0}

# Layer 3 — coerce blanks into real NaN so they can be counted
df["totalcharges"] = pd.to_numeric(df["totalcharges"].astype(str).str.strip(),
                                   errors="coerce")
# -> 11 NaN, dtype float64
```

The 11 affected customers were then profiled before any fix was chosen. **All 11 have tenure = 0** — they joined in the current billing period and have not been charged yet. That single fact is what makes the correct fill value obvious.

Diagnostic	Result
Rows with missing totalcharges	11 (0.16% of 7,043)
Tenure of those rows	**0 months for all 11** (verified programmatically)
Example	customerid 4472-LVYGI · tenure 0 · monthlycharges $52.55 · Two year contract
Median totalcharges (reference only)	$1,397.47

Decision: fill with 0, and why the alternatives were rejected

Option	Value used	Verdict
A. Fill with 0 — **chosen**	$0.00	**Business-correct.** 0 months of tenure × a monthly plan = $0 billed so far. Keeps all 7,043 customers and keeps the distribution honest.
B. Drop the 11 rows	—	Technically safe (0.16% loss) but discards valid, paying customers for no analytical gain.
C. Fill with median ($1,397.47)	$1,397.47	**Wrong.** Assigns roughly a year of charges to customers who have never been billed, inflating average lifetime value and flattening the right skew.
D. Fill with mean ($2,283.36)	$2,283.36	**Worse than C.** Even more distortion, and sensitive to the long tail.

> **Documented reasoning**
>
> The brief permits "fill, drop, or impute". Fill with zero was selected because it is the only option that is simultaneously lossless, reproducible and arithmetically correct for these rows. The alternatives are recorded above so a reviewer can see the trade-off was considered rather than defaulted.

```
df["totalcharges"] = df["totalcharges"].fillna(0)   # tenure = 0  ->  $0 billed
```

4.4 Step 3 — Duplicate removal

Duplicates would double-count customers and inflate every rate and chart downstream. The check was run at two levels before drop_duplicates() was applied.

Check	Result	Action
Full-row duplicates	0	drop_duplicates() applied (idempotent)
customerid unique values	7,043 of 7,043 rows	No ID collisions — no dedup by key required
Rows before / after	7,043 → 7,043	Confirmed nothing was removed

4.5 Final hygiene and validation

A last safety pass strips whitespace from every text column — a stray trailing space would split a group into "Yes" and "Yes " and quietly corrupt every demographic breakdown in Tasks 3–5. The cleaned frame is then written to disk and re-read to prove the file on disk is valid.

Validation check	Before cleaning	After cleaning	Result
Shape	(7043, 21)	(7043, 21)	PASS — no rows lost
True nulls (isnull)	0	0	PASS
Hidden blank strings	11 in TotalCharges	0	PASS
totalcharges dtype	object (text)	float64	PASS
Duplicate rows	0	0	PASS
Column names	Mixed case	21 lowercase, no spaces, unique	PASS

Deliverable: Telco_Customer_Churn_Cleaned.csv — 7,043 rows × 21 columns, 0 nulls, 0 duplicates, numeric totalcharges. Tasks 3 and 4 both load this file, and both fall back to re-applying the Task 2 fix if it is missing, so each notebook remains runnable on its own.

4.6 Findings

- **No data was discarded.** Shape is unchanged at (7,043, 21) — correct, because the only real defects were naming and hidden blanks, neither of which requires dropping rows.
- **The one substantive repair was totalcharges:** text with 11 blanks → float64 with zero nulls, filled with 0 on business grounds.
- **Naming is now consistent** across all 21 columns, verified by assertions rather than by eye.
- **The dataset is provably analysis-ready:** 0 nulls, 0 duplicates, unique business key, correct dtypes.
- **Every cleaning decision is reversible and justified** — the rejected alternatives are recorded rather than silently discarded.

5. Task 3 — Exploratory Data Analysis

Objective: understand the trends and distributions in the data through summary statistics, numerical-column visualisations and churn-rate analysis.

5.1 Summary statistics

describe() on the three numerical columns establishes the centre, spread and shape of every quantitative attribute before any chart is drawn.

Statistic	tenure (months)	monthlycharges ($)	totalcharges ($)
count	7,043	7,043	7,043
mean	32.37	64.76	2,279.73
std	24.56	30.09	2,266.79
min	0	18.25	0.00
25% (Q1)	9	35.50	398.55
50% (median)	29	70.35	1,394.55
75% (Q3)	55	89.85	3,786.60
max	72	118.75	8,684.80
skewness	0.24	−0.22	0.96

Column	Mean	Median	Mode	Reading
tenure	32.37	29	1	Wide spread with spikes at brand-new (1 month) and loyal (72 months)
monthlycharges	64.76	70.35	20.05	Mean below median → a dense low-price cluster pulls the average down
totalcharges	2,279.73	1,394.55	0.00	Mean well above median, skew 0.96 → strongly right-skewed lifetime value

seniorcitizen is a 0/1 flag, so its mean is the share: **16.21%** of customers are senior citizens.

Modal categorical profile: male · no partner · no dependents · month-to-month contract · fiber optic · electronic check · no device protection · **did not churn**.

5.2 Distributions of the numerical columns

Three histograms with kernel-density overlays show the shape of each numerical attribute.

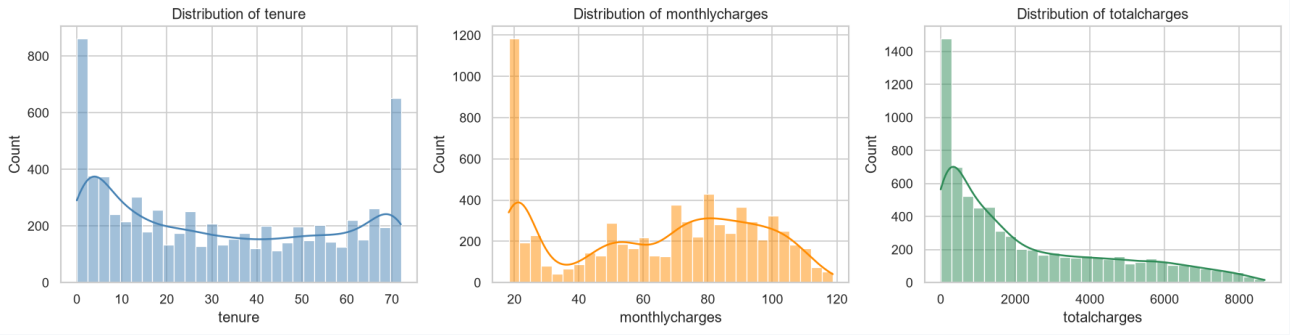


Figure 1. Distribution of the three numerical columns. **tenure** is broadly spread but piled up at both ends — a large spike of brand-new customers at 1 month and a second rise at the 72-month ceiling. **monthlycharges** is distinctly bimodal, with a tall spike near $20 (basic DSL-only plans) and a broad second hump around $70–$105 (fiber plus add-ons). **totalcharges** is strongly right-skewed: most customers have paid a modest cumulative amount, with a long tail of high-lifetime-value accounts running out towards $8,685.

5.3 Spread and outliers, split by churn

Box plots show median, interquartile range and outliers, and splitting by the churn flag makes the central EDA comparison visible immediately.

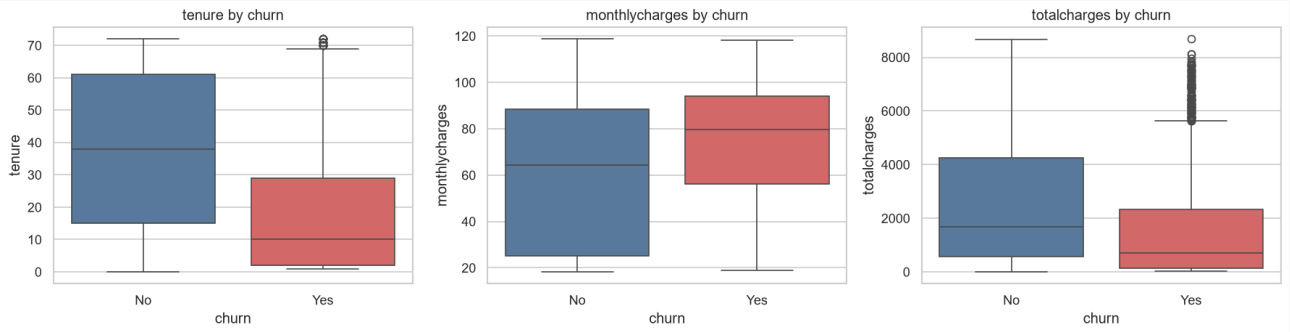


Figure 2. Box plots of each numerical column split by churn status. The **tenure** panel is the striking one: the median tenure of churners sits far below that of stayers. The **monthlycharges** panel runs the other way — churners are concentrated at the higher end of the price range.

Group	Customers	Avg tenure (months)	Avg monthlycharges ($)	Avg totalcharges ($)
Stayed (churn = No)	5,174	37.57	61.27	2,549.91
Churned (churn = Yes)	1,869	17.98	74.44	1,531.80
Difference	—	**−19.59 months**	**+13.17 (+21.5%)**	**−1,018.11 (−39.9%)**

The key tension in the data

Churners pay **$13.17 more per month** than customers who stay, but they leave roughly **19.6 months earlier**, so they generate about **40% less lifetime value**. Higher bills are therefore associated with churn not because price causes churn, but because the higher-priced plans — fiber with add-ons — are where the dissatisfaction concentrates. This is the pattern the segmentation in Task 4 makes explicit.

5.4 Churn rate — overall

Churn status	Customers	Share of base
No (retained)	5,174	73.46%
Yes (churned)	1,869	**26.54%**
Total	**7,043**	**100.00%**

Roughly **one in four customers churns**. The class is imbalanced but the minority is large enough (1,869 records) for segment-level analysis to be statistically meaningful.

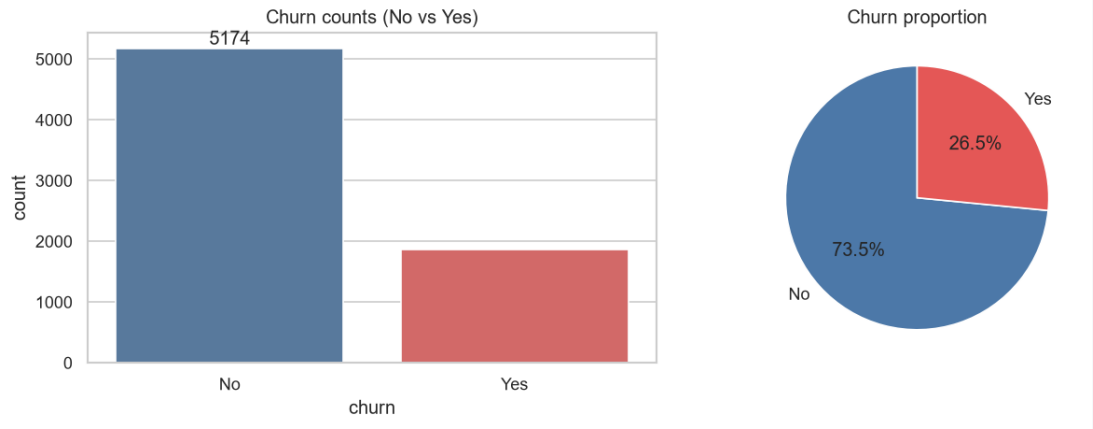


Figure 3. Churn counts and proportions. The bar chart shows absolute volumes (5,174 retained vs 1,869 churned) while the pie chart shows the same split as a share of the base (73.46% / 26.54%).

5.5 Where the churn sits — breakdowns by contract, service and payment

The overall 26.54% hides very uneven pockets. Row-normalised crosstabs express every group as a percentage of itself, so groups of different sizes can be compared directly.

Dimension	Segment	Churn rate	Read
Contract	Month-to-month	**42.71%**	Baseline risk — no commitment
	One year	11.27%	Risk falls by ~¾
	Two year	**2.83%**	Effectively locked in — 15× safer than month-to-month
Internet service	Fiber optic	**41.89%**	Highest-risk product
	DSL	18.96%	Mature, stable product
	No internet service	7.40%	Lowest churn of all groups
Payment method	Electronic check	**45.29%**	Manual payment = highest friction
	Bank transfer (automatic)	16.71%	Auto-pay more than halves churn
	Credit card (automatic)	15.24%	Best-performing payment method
	Mailed check	19.11%	Manual, but low-value segment

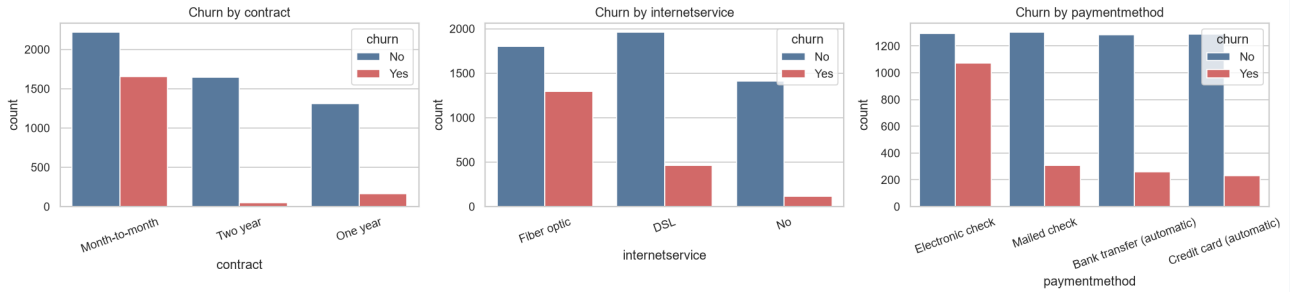


Figure 4. Churn composition by contract, internet service and payment method. Each panel counts customers split by churn status within the category. Month-to-month and fiber optic contribute the large red blocks; two-year contracts are almost entirely blue.

5.6 Findings

- **tenure:** mean 32.4, median 29.0, mode 1 month, range 0–72 — a wide spread with spikes at both ends of the lifecycle.
- **monthlycharges:** mean $64.76, median $70.35, range $18.25–$118.75 — bimodal, with a basic-plan cluster near $20 and a fiber cluster near $80–$105.
- **totalcharges:** mean $2,279.73 against a median of $1,394.55 (skew 0.96) — a small number of customers carry a large share of lifetime value. The mode of $0.00 is the honest fingerprint of the 11 tenure-0 customers repaired in Task 2.
- **Churn overall:** 26.54% (1,869 of 7,043).
- **Churners leave early and pay more:** 17.98 vs 37.57 months of tenure, but $74.44 vs $61.27 per month.
- **Three hotspots identified:** month-to-month (42.71%), electronic check (45.29%) and fiber optic (41.89%). Two-year contracts (2.83%) and automatic payment (15–17%) are the safe zones.

6. Task 4 — Customer Segmentation Visualization

Objective: visualise the customer distribution by tenure and compare average monthly charges across tenure categories with annotated charts, using both static and interactive tooling.

6.1 How the segments were defined

Tenure is continuous from 0 to 72 months, but the business reasons in lifecycle stages. The brief specifies the three buckets, and pd.cut reproduces them exactly:

```
df["tenure_group"] = pd.cut(df["tenure"], bins=[-1, 12, 36, 72],
                            labels=["0-12 months", "13-36 months", "37+ months"])
```

Segment	Business meaning	Boundary rule	Customers	Share
0–12 months	New / onboarding	tenure ≤ 12	2,186	31.04%
13–36 months	Developing	13 ≤ tenure ≤ 36	1,856	26.35%
37+ months	Loyal	tenure ≥ 37	3,001	42.61%
Total	—	0–72 months	**7,043**	**100.00%**

The bucket counts are asserted to sum to the full dataset (3,001 + 2,186 + 1,856 = 7,043), so no customer is dropped or double-counted by the binning.

6.2 Customer distribution by tenure — donut chart

The brief asks for the tenure distribution as a pie or donut chart. The base is loyal-leaning — 42.61% have been customers for more than three years — but nearly a third (31.04%) are still in their first year, and Task 3 showed that is precisely where

churn concentrates. The donut form is used because the empty centre carries the sample size, and because three slices with clearly unequal angles are easier to read than an area-based pie.

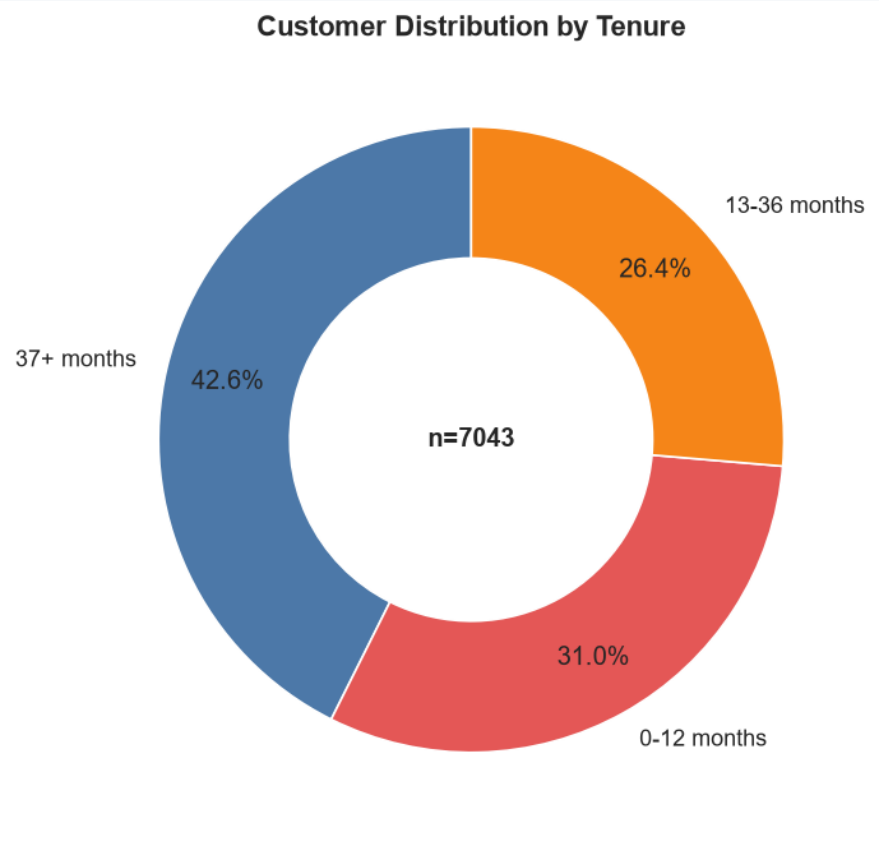


Figure 5. Customer distribution by tenure segment (Matplotlib donut). Loyal customers (37+ months) form the largest slice at 42.6%, but the 0–12 month newcomer slice at 31.0% is the segment the retention strategy must act on. Segment sizes are labelled as percentages with the total sample size shown in the centre.

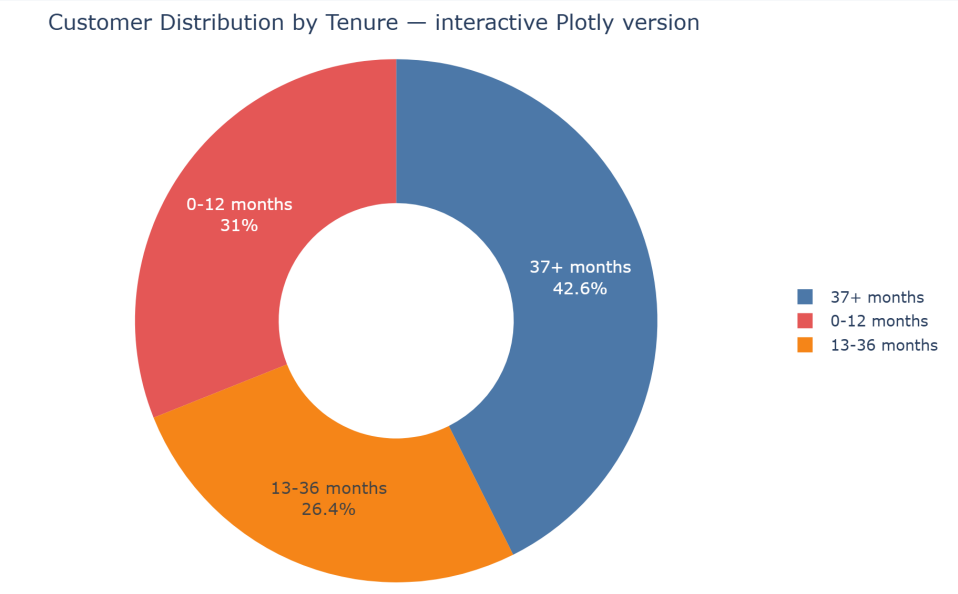


Figure 6. The same distribution as an **interactive** Plotly donut — hovering a slice reveals the customer count alongside the percentage, and clicking a legend entry isolates that segment. This satisfies the brief's interactive visualization requirement; the live version is in the notebook.

6.3 Average monthly charges by tenure — clustered bar chart

The second requirement is a clustered bar chart of average monthly charges across tenure categories, annotated to highlight the significant trend. Bars are clustered by churn status so the value comparison and the risk comparison are read from a single chart.

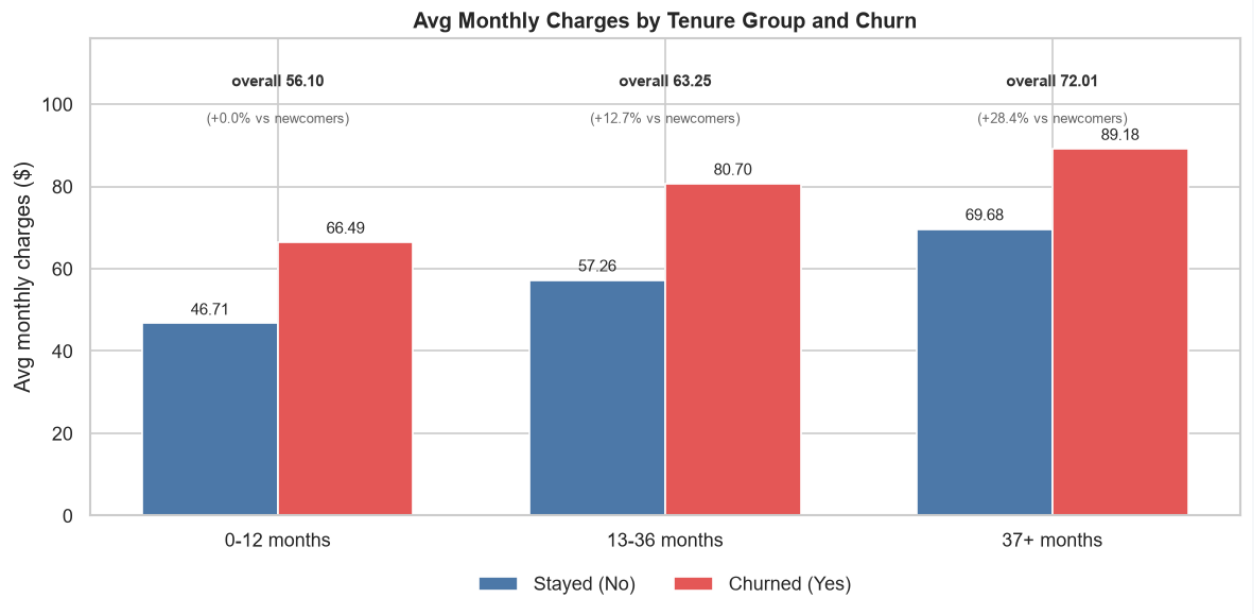


Figure 7. Average monthly charges by tenure group, clustered by churn status (Matplotlib). Every bar carries its exact value. Above each cluster the overall average for that tenure group is annotated, with the percentage difference against newcomers — the rising **$56.10 → $63.25 → $72.01** trend the brief asks to be highlighted.

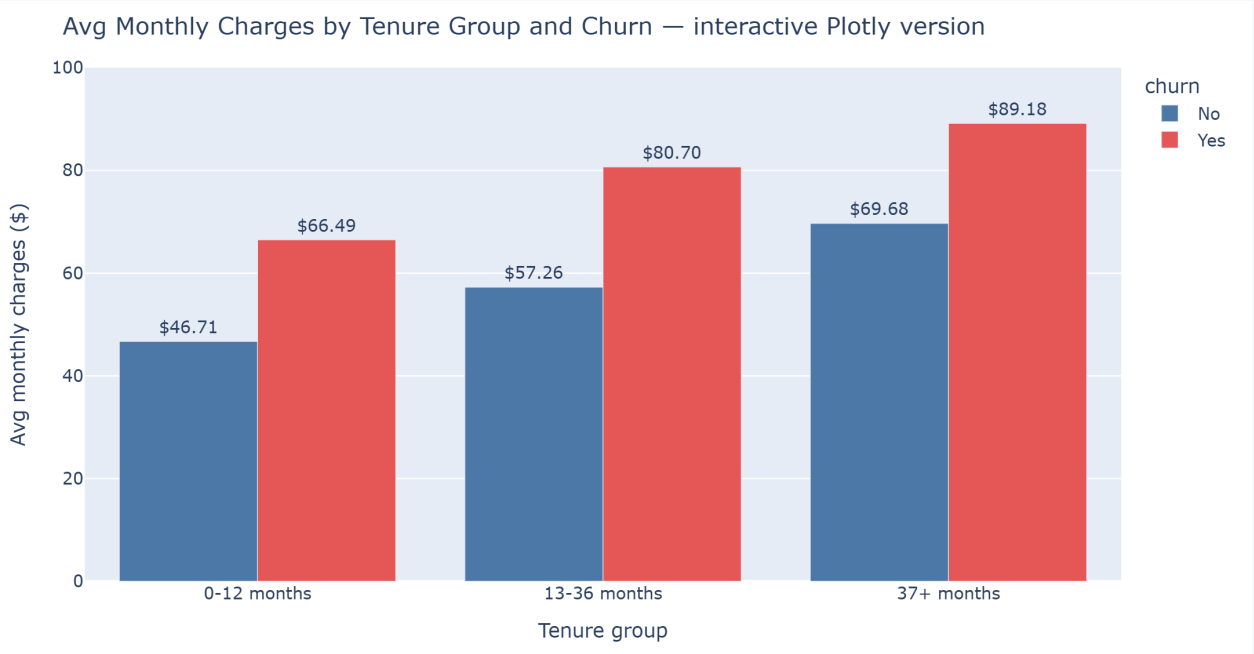


Figure 8. Interactive Plotly version of the clustered bar chart. Hovering any bar returns the exact dollar value, and the churn series can be toggled from the legend to compare stayers and churners independently.

Tenure group	Customers	Stayed — No	Churned — Yes	Gap	Overall average	vs newcomers
0–12 months	2,186	$46.71	$66.49	+$19.78	**$56.10**	baseline
13–36 months	1,856	$57.26	$80.70	+$23.44	**$63.25**	+12.7%
37+ months	3,001	$69.68	$89.18	+$19.50	**$72.01**	+28.4%

What the annotations reveal

1. Value rises with tenure. The overall average climbs from $56.10 to $72.01, a 28.4% uplift. Long-tenure customers are not legacy discount accounts — they are the highest-value segment in the business, consistent with gradual upselling to fiber and add-on plans.

2. Churners out-pay stayers in every single bucket. The gap is +$19.78, +$23.44 and +$19.50 respectively — not a first-year phenomenon but a structural one. Price sensitivity is a whole-book issue, not just a newcomer issue.

6.4 Findings

- **Segment mix:** 42.61% loyal (37+ months), 31.04% newcomers (0–12), 26.35% developing (13–36) — a loyal core carrying a heavy, fragile newcomer tail.
- **Value trend:** average monthly bill rises monotonically with tenure, $56.10 → $63.25 → $72.01 (+28.4%). The loyal segment is the highest-ARPU segment.
- **Risk overlay:** within every tenure bucket, churners pay more than stayers — so the retention conversation is not simply about discounting cheap customers.
- **Brief requirements met:** tenure distribution as a donut (static + interactive) □ · clustered bar of average monthly charges by tenure □ · annotations highlighting the significant trend □ · interactive visualisation with Plotly □.
- **Handed to Task 5:** the three segments become the grouping key for charge and churn statistics, and the churn-versus-value contrast becomes the core retention narrative.

7. Consolidated Findings & Recommendations

Everything Tasks 1–4 established, and the actions the evidence supports.

7.1 Findings

#	Finding	Evidence (Tasks 1–4)
F1	The raw dataset carried hidden missing data that the standard check missed	0 true nulls, but 11 blank totalcharges strings, all with tenure = 0 (Task 1)
F2	The dataset can be cleaned losslessly	7,043 → 7,043 rows; 0 nulls, 0 duplicates, totalcharges now float64 (Task 2)
F3	Roughly one customer in four churns	26.54% — 1,869 of 7,043 (Task 3)
F4	Churn is overwhelmingly a first-year problem	47.44% churn at 0–12 months vs 11.93% at 37+ months (Tasks 3 & 4)
F5	Churners pay more per month than customers who stay	$74.44 vs $61.27 (+21.5%), in every tenure bucket (Tasks 3 & 4)
F6	Churners generate about 40% less lifetime value	$1,531.80 vs $2,549.91 average total charges (Task 3)

#	Finding	Evidence (Tasks 1–4)
F7	Customer value rises with tenure, so early saves compound	Average monthly $56.10 → $63.25 → $72.01 (+28.4%) (Task 4)
F8	Contract commitment is the strongest churn shield	Month-to-month 42.71% vs one year 11.27% vs two year 2.83% (Task 3)
F9	Manual payment methods signal dissatisfaction	Electronic check 45.29% vs automatic 15–17% (Task 3)
F10	Fiber optic carries disproportionate risk	41.89% churn vs DSL 18.96% and no-internet 7.40% (Task 3)

7.2 Recommended actions

The recommendations below are ordered by the size of the churn pool they address. Expected effects are directional statements based on the observed segment differences, not forecasts from a fitted model.

Priority	Recommended action	Rationale from Tasks 1–4	Population affected
1 — High	Launch a first-90-days onboarding programme for new customers: early technical check-ins, service-quality review and a clear path to a better-priced plan.	The 0–12 month segment is 31.04% of the base (2,186 customers) and churns at 47.44% — more than four times the loyal-segment rate. No other single pool offers this much upside.	2,186 customers (~1,037 churn events)
2 — High	Drive a move from month-to-month to a one- or two-year contract, using a targeted introductory discount on the first year of an annual plan.	Contract type is the strongest lever in the data: 42.71% vs 11.27% vs 2.83%. Closing even part of that gap is worth more than any acquisition campaign.	3,255 month-to-month customers
3 — Medium	Move electronic-check customers onto automatic payment with a small prompt-time incentive.	Electronic check churns at 45.29% against 15.24–16.71% for the automatic methods — the single highest-churn payment group and the easiest to convert.	2,875 electronic-check customers
4 — Medium	Introduce proactive service reviews for fiber-optic customers in months 1–9, before the churn spike, and review fiber pricing and installation quality.	Fiber optic churns at 41.89% against 18.96% for DSL. Fiber customers also pay the highest monthly charges, so this is where revenue per lost customer is highest.	3,020 fiber-optic customers
5 — Medium	Build a value-recovery offer for high-billing churners detected before cancellation (retention discount, plan optimisation, priority support).	Churners pay $13.17 more per month than stayers in every tenure bucket — the revenue at risk per departure is above average, and a save is worth more here.	1,869 churners (pre-churn trigger)
6 — Ongoing	Institutionalise the data-quality checks from Tasks 1–2 in the ingestion pipeline, in particular the hidden-blank test that isnull() alone would miss.	11 blank totalcharges values were invisible to the standard missing-value check. The same blind spot will recur on every future extract.	All future loads

7.3 Measurement

Once Task 5 and a predictive model are in place, the actions above should be tracked with three metrics that this analysis has already defined: overall churn rate (baseline 26.54%), churn rate within the 0–12 month segment (baseline 47.44%), and churn rate among month-to-month contract holders (baseline 42.71%). Reporting churn alongside average monthly charges keeps the trade-off visible, because a retention action that lowers churn while shifting customers to cheaper plans may not improve value at all.

8. Conclusion

Tasks 1 to 4 converted a raw CSV into an analysis-ready dataset and a defensible picture of where churn comes from. The technical work is complete and reproducible: 7,043 rows in, 7,043 rows out, no nulls, no duplicates, every column name verified, every cleaning decision documented with its rejected alternatives.

The analytical conclusion is sharper than a headline churn number. Churn in this business is not spread evenly across the customer base — it concentrates in the first year, in month-to-month contracts, among manual-payment customers and on the fiber product, and it costs more per customer than the average because those customers pay higher monthly bills. Meanwhile the customers who survive become progressively more valuable, so retention is not a cost centre: the average monthly bill rises 28.4% between the newest and the longest-tenured segment.

That combination — concentrated, addressable early-stage leakage feeding a rising value curve — is what makes this a tractable retention problem. The segmentation built in Task 4 gives the operational teams a common language for it: **new**, **developing** and **loyal**. Task 5 quantifies the same segments by demographics, contract and payment method and turns the patterns above into lifecycle recommendations.

Deliverables accompanying this report

Notebooks: Task_1_Understand_Dataset.ipynb · Task 2 - Data Cleaning.ipynb · Task 3 - Exploratory Data Analysis.ipynb · Task 4 - Customer Segmentation Visualization.ipynb
Dataset: Telco_Customer_Churn_Cleaned.csv (produced by Task 2)
Source: Telco_Customer_Churn_Dataset (3).csv (unmodified)

pendix Reproducibility Notes

A.1 How to reproduce this report

1. Place Telco_Customer_Churn_Dataset (3).csv in the project folder.
2. Run the notebooks in order: Task 1 → Task 2 → Task 3 → Task 4. Task 2 writes Telco_Customer_Churn_Cleaned.csv; Tasks 3 and 4 read it.
3. Tasks 3 and 4 each fall back to re-applying the Task 2 cleaning logic if the cleaned file is absent, so either notebook also runs standalone.
4. Every notebook was executed end-to-end with **zero errors and zero warnings**; all outputs and figures in this report are taken directly from those runs.

A.2 Environment

Component	Version
OS	Windows
Python	3.14.7
pandas	3.0.6
NumPy	2.5.3
Matplotlib	3.11.2
Seaborn	0.13.2
Plotly	7.1.0
Kernel	python3 (ipykernel 7.4.0)

A.3 File inventory

File	Type	Contents
BUSINESS ANALYSIS TASK LIST (1).pdf	Brief	Internship task list and submission instructions
Telco_Customer_Churn_Dataset (3).csv	Data (raw)	7,043 × 21 — unmodified source
Telco_Customer_Churn_Cleaned.csv	Data (clean)	7,043 × 21 — Task 2 output
Task_1_Understand_Dataset.ipynb	Notebook	12 cells (5 code, 7 markdown)
Task 2 - Data Cleaning.ipynb	Notebook	20 cells (10 code, 10 markdown)
Task 3 - Exploratory Data Analysis.ipynb	Notebook	16 cells (7 code, 9 markdown), 4 figures
Task 4 - Customer Segmentation Visualization.ipynb	Notebook	14 cells (6 code, 8 markdown), 4 charts (2 static + 2 interactive)

A.4 Validation summary

The checks below were re-run against the final cleaned dataset immediately before this report was generated.

Check	Expected	Observed	Status
Rows	7,043	7,043	PASS
Columns	21	21	PASS
Total nulls	0	0	PASS
Duplicate rows	0	0	PASS
totalcharges dtype	float64	float64	PASS
totalcharges zeros (repaired)	11	11	PASS
Tenure segment total	7,043	7,043	PASS
Churn split (No / Yes)	5,174 / 1,869	5,174 / 1,869	PASS

End of report — Tasks 1 to 4. Task 5 (Advanced Analysis) is delivered separately.